

Final Year Project Long Proposal

FYPHub: An Agentic AI-Driven Platform for Intelligent Final Year Project Management

Muhammad Maaz 23108179

Zain Ali Kiani 23108197

Danish Faraz 23108160

Supervised by: Sir Hamza Imran

Department of Robotics & Artificial Intelligence
Faculty of Computing and Engineering Sciences
SZABIST University
Islamabad, Pakistan

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Revision History

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Muhammad Maaz	Sir Hamza Imran	07-Oct-2026	Long proposal structure and content	1.0
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1. Project Stream

- [x] Web-Based FYP
- [ ] Desktop Application
- [ ] Mobile Application
- [ ] Game
- [ ] Hardware-Based

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2. Introduction

Final Year Project (FYP) management is a multi-stage academic process involving proposal processing, project allocation, collaboration, project completion, evaluation, and administrative coordination. Prior FYP-management research has shown that these activities benefit from centralized digital workflows and role-specific collaboration tools rather than disconnected manual processes [1, 2]. In many university environments, forms, spreadsheets, emails, messaging applications, and loosely organized digital records can make proposal tracking, supervisor coordination, evaluation, and reuse of previous project knowledge difficult.

FYPHub is proposed as a centralized, role-aware, AI-assisted platform for managing the complete FYP lifecycle. The platform is designed for five major roles: Student, Supervisor, FYP Coordinator, Panel Member, and Administrator. It combines deterministic academic rules with Large Language Models (LLMs), semantic embeddings, Retrieval-Augmented Generation (RAG), GraphRAG, machine-learning based risk analysis, optimization techniques, and agentic AI orchestration. RAG provides language models with retrieved external evidence for knowledge-intensive generation, while GraphRAG extends retrieval with graph-structured relationships for broader and multi-hop reasoning over document collections [3, 4]. AI is used for assistance, analysis, retrieval, explanation, and prediction, while final academic authority remains with authorized university personnel.

The FYP is organized into twelve clearly separated modules. Each module owns a distinct part of the FYP lifecycle so that proposal management, quality compliance, originality checking, supervision, scheduling, evaluation, progress monitoring, institutional knowledge retrieval, and final archiving do not overlap unnecessarily.

3. Project Description

FYPHub is a responsive web-based platform that digitizes and intelligently connects the complete Final Year Project process. Students can form groups, find compatible teammates, create and submit proposal versions, run AI-assisted compliance checks, compare proposals against previous FYPs, and request suitable supervisors. Supervisors can review proposals using structured comments and revision requests, while the Coordinator manages academic routing and panel readiness. The system also schedules defenses, supports structured panel evaluation, tracks project progress and supervision evidence, archives completed FYPs, and makes institutional project knowledge searchable through RAG and GraphRAG.

The platform follows a shared-module, role-based-access architecture. A feature is not duplicated into separate modules merely because different roles can view it. For example, the proposal compliance engine remains one module, but students, supervisors, coordinators, and panel members access different views according to their permissions. Similarly, the final archive creates and preserves institutional knowledge, while the knowledge-search module retrieves and reasons over that archived information.

The system is expected to provide a single source of truth for proposal versions,

supervisor assignments, review comments, coordinator actions, panel evaluations, project progress, and final project outcomes. This improves traceability and supports data-driven academic decision making across the entire FYP lifecycle.

4. Real World Problem

Universities generate a large amount of valuable FYP knowledge every semester, but that knowledge is often difficult to search, compare, and reuse. Students may not know whether an idea has already been implemented, what limitations previous projects had, which supervisor is most suitable, or which university formatting and proposal requirements are still unmet. Group formation can also be difficult when students do not have suitable teammates, and duplicate membership can occur if group rules are not centrally enforced.

FYP administration also involves several academic actors. Students submit proposals, supervisors review them, coordinators control academic routing, panels evaluate the work, and administrators configure academic rules and deadlines. When these activities are distributed across disconnected tools, important context is lost. Revision comments can become scattered, supervisor capacity may be unclear, panel scheduling can create conflicts, progress evidence may remain informal, and completed projects can become static files rather than reusable institutional knowledge.

The real-world challenge is therefore broader than simple proposal submission. The university needs an integrated system that manages the workflow, applies deterministic academic rules, provides AI-assisted analysis where appropriate, preserves an auditable history, and converts historical FYP records into an active knowledge base for future students and faculty.

5. Problem Statement

Current FYP management practices may suffer from fragmented team formation, manual proposal submission, inconsistent formatting, weak originality checks, limited visibility of supervisor expertise and capacity, scattered review communication, manual defense scheduling, disconnected panel feedback, limited progress evidence, and poor reuse of completed-project knowledge. Existing generic learning-management, plagiarism-detection, and project-management tools can support individual parts of this process, but they do not provide a unified FYP-specific platform that connects academic workflow, proposal intelligence, supervisor allocation, panel evaluation, project progress, and institutional knowledge.

The proposed project aims to develop the **FYPHub** platform, which manages the complete FYP lifecycle through twelve integrated but non-overlapping modules. The system will combine role-based workflow control with AI-assisted proposal compliance, semantic originality analysis, supervisor matching, revision verification, scheduling optimization, progress and risk intelligence, RAG-based institutional knowledge search, and final project handover support.

6. Names of Existing Similar FYPs (if repeated)

No exact repeated FYP has been confirmed in the current project records at proposal stage. However, the problem space overlaps partially with learning-management systems, plagiarism/similarity platforms, project-management tools, and institutional repositories. The proposed FYPHub differs by integrating the complete FYP lifecycle and by using archived institutional FYP data as an active source for similarity analysis, knowledge retrieval, extension guidance, and analytics.

The proposed system will also contain its own **Proposal Similarity & Originality Check** module. Once historical FYP records are available in the repository, this module will compare a new proposal against previous projects and provide evidence of semantic, functional, and methodological overlap rather than relying only on title similarity.

7. New Functionalities (main) to be added Features

FYPHub introduces several capabilities that are not normally available together in a single FYP management platform:

- AI-assisted teammate discovery using compatibility questions, skills, availability, work style, and role preferences, with duplicate-group membership prevention.
- University-template-aware proposal compliance checking for structure, headings, margins, spacing, alignment, required sections, and academic consistency.
- Semantic and functional comparison of new proposals against archived FYPs, including contribution boundaries and extension gaps.
- Expertise, workload, preference, and capacity-based supervisor recommendation and allocation with slot enforcement.
- Structured supervisor comments, revision requests, two-way discussion, and AI-assisted verification of whether requested revisions were resolved.
- Coordinator-controlled academic routing with delay and workflow-conformance monitoring.
- Conflict-aware defense scheduling and panel assignment based on availability, room/time constraints, and academic conflicts.
- Panel scoring and feedback with AI-generated viva questions, evaluator disagreement detection, and feedback-resolution tracking.
- Supervision-meeting evidence, milestone tracking, project-health classification, delay-risk analysis, and recovery recommendations.
- Final FYP archiving with implemented-scope extraction, limitations, future work, reusable assets, and intelligent handover packages.
- RAG/GraphRAG-based institutional FYP search with evidence and multi-hop reasoning over archived projects.
- Institutional analytics for domain trends, workload, turnaround time, completion patterns, and management insights.

8. Application Review

This section reviews six existing platforms that address selected parts of the FYP lifecycle. Each application is summarized through two key characteristics and two limitations, followed by a feature-by-application comparison table in the same tabular pattern used throughout this proposal. The purpose is to show which capabilities already exist in established systems and which FYP-specific gaps remain.

8.1 Moodle Assignment Activity [5]

Moodle provides structured assignment submission, group submissions, deadlines, grading, feedback, and rubric-based assessment within a learning-management environment.

- **Key characteristic 1:** Students can submit files or online text individually or in groups, with configurable availability and due dates.
- **Key characteristic 2:** Instructors can grade submissions, provide feedback, and use rubrics or custom grading scales.

The limitations of Moodle for the target FYP lifecycle are listed below.

- **Limitation 1:** It does not provide FYP-specific supervisor expertise/capacity matching, allocation optimization, or defense-panel scheduling.
- **Limitation 2:** It does not provide institutional FYP semantic originality analysis, GraphRAG reasoning, or intelligent handover/continuation support.

8.2 Turnitin Similarity Report [6]

Turnitin compares submitted text against large source collections and produces a Similarity Report that highlights matching text, source information, and an overall similarity percentage.

- **Key characteristic 1:** Text matching against internet sources, publications, and previously submitted work with source-level evidence.
- **Key characteristic 2:** Similarity reports support academic review by grouping and presenting matched passages for human interpretation.

The limitations of Turnitin for the target FYP lifecycle are listed below.

- **Limitation 1:** It primarily measures textual similarity and does not evaluate FYP problem, objectives, modules/features, methodology, implemented scope, and contribution boundaries as one semantic project comparison.
- **Limitation 2:** It does not manage FYP team formation, supervisor allocation, coordinator routing, defense scheduling, project-progress intelligence, or final handover.

8.3 OpenProject [7]

OpenProject is a general project-management and collaboration platform supporting planning, tasks, milestones, workloads, timelines, and project progress.

- **Key characteristic 1:** Work packages, tasks, milestones, timelines, and responsibilities can be planned and tracked across a project lifecycle.
- **Key characteristic 2:** Team collaboration, workload planning, project dashboards, and status tracking support coordinated project delivery.

The limitations of OpenProject for the target FYP lifecycle are listed below.

- **Limitation 1:** It does not provide university-template proposal compliance, historical FYP originality intelligence, or formal supervisor revision-compliance checking.
- **Limitation 2:** It does not provide FYP-specific supervisor allocation, panel evaluation, institutional FYP archive reasoning, or GraphRAG-based knowledge search.

8.4 Microsoft Teams for Education [8]

Microsoft Teams for Education combines class communication, channels, assignments, due dates, resources, feedback, and rubric-based grading in one collaborative environment.

- **Key characteristic 1:** Channels and class-team communication support collaboration among students and educators.
- **Key characteristic 2:** Assignments support due dates, file submission, resubmission, resources, grades, and rubrics.

The limitations of Microsoft Teams for Education for the target FYP lifecycle are listed below.

- **Limitation 1:** It does not provide FYP-specific teammate compatibility matching, supervisor expertise/capacity allocation, or historical FYP originality analysis.
- **Limitation 2:** It does not manage coordinator academic routing, defense-panel optimization, project handover intelligence, or RAG/GraphRAG reasoning over FYP archives.

8.5 GitHub Projects [9]

GitHub Projects supports software-team planning and tracking through issues, pull requests, project boards, tables, roadmaps, custom fields, and code-linked work items.

- **Key characteristic 1:** Issues and Projects can organize tasks, ideas, bugs, milestones, and development work with configurable views and metadata.
- **Key characteristic 2:** Work tracking is closely linked with repositories, pull requests, and software-development activity.

The limitations of GitHub Projects for the target FYP lifecycle are listed below.

- **Limitation 1:** It is development-work oriented and does not provide university proposal formatting/compliance, supervisor matching, or formal academic routing.
- **Limitation 2:** It does not provide defense scheduling, panel rubric evaluation, semantic comparison with archived FYPs, or institutional FYP handover workflows.

8.6 DSpace Institutional Repository [10]

DSpace is an institutional repository platform for managing, preserving, controlling access to, and searching digital content and metadata.

- **Key characteristic 1:** It preserves multiple digital formats and supports metadata/full-text search for institutional collections.
- **Key characteristic 2:** Granular access control and repository organization support long-term institutional knowledge preservation.

The limitations of DSpace for the target FYP lifecycle are listed below.

- **Limitation 1:** It is primarily a repository and does not manage active FYP team formation, proposal review, supervisor allocation, progress, or defense workflows.
- **Limitation 2:** Standard repository search does not provide FYP-specific semantic originality analysis, revision verification, or RAG/GraphRAG-based multi-hop academic reasoning by itself.

8.7 FYPHub (Our FYP)

FYPHub combines the complete university FYP lifecycle in one role-aware platform instead of treating submission, similarity checking, supervision, evaluation, progress, and institutional knowledge as disconnected activities.

- **Key characteristic 1:** Integrated 12-module workflow covering team formation, proposal compliance/originality, supervisor allocation/revision, coordinator control, defense/panel processes, project progress, final archive, and handover.
- **Key characteristic 2:** AI-assisted institutional intelligence using LLMs, embeddings, RAG/GraphRAG, optimization, and evidence-based analysis while retaining human academic authority.

Table 1: Applications Comparison

Feature	**Moodle [5]**	**Turnitin [6]**	**OpenProject [7]**	**MS Teams [8]**	**GitHub [9]**	**DSpace [10]**	**FYPHub**
Assignment/document submission and deadlines	✓	✓	✗	✓	✗	✗	✓
Team collaboration and work tracking	✗	✗	✓	✓	✓	✗	✓
Similarity/originality evidence	✗	✓	✗	✗	✗	✗	✓
Rubric-based evaluation/feedback	✓	✓	✗	✓	✗	✗	✓
Institutional repository/archive preservation	✗	✗	✗	✗	✗	✓	✓
University proposal format/compliance analysis	✗	✗	✗	✗	✗	✗	✓
Supervisor expertise/capacity allocation	✗	✗	✗	✗	✗	✗	✓
Historical FYP semantic/functional comparison	✗	✗	✗	✗	✗	✗	✓
Defense scheduling and panel assignment	✗	✗	✗	✗	✗	✗	✓
RAG/GraphRAG reasoning over archived FYPs	✗	✗	✗	✗	✗	✗	✓
Final FYP handover/continuation intelligence	✗	✗	✗	✗	✗	✗	✓

9. Scope

The scope of FYPHub covers the complete university FYP lifecycle through twelve integrated modules. The system is software-based and uses role-aware workflows, deterministic academic rules, and AI-assisted analysis while preserving human academic authority.

9.1 Included Aspects

- Role-based web access for Student, Supervisor, FYP Coordinator, Panel Member, and Administrator users.

- Student team formation, proposal submission/versioning, university-format compliance, and proposal originality analysis.
- Supervisor matching/allocation, supervisor-led proposal review/revision, coordinator routing, defense scheduling, and panel evaluation.
- Supervision evidence, milestone/progress monitoring, project-risk assistance, and recovery recommendations.
- Final FYP archiving, handover/continuation knowledge, RAG/GraphRAG search, and institutional analytics.

9.2 Excluded Aspects

- Automatic replacement of supervisor, coordinator, panel, or administrator academic decisions by AI; authorized personnel retain final authority.
- University processes outside the FYP lifecycle such as general admissions, fee processing, and non-FYP academic administration.
- Development of proprietary hardware or a native mobile application; the proposed system is a web-based software platform.
- Use of confidential institutional or student data outside approved access permissions and role-based policies.

9.3 Modules Description

Module 1: Admin Dashboard & Academic Management

Description: This module provides the administrative foundation of the platform. It allows authorized administrators to manage users, roles, academic cycles, proposal templates, supervisor capacity, evaluation rubrics, deadlines, reminders, escalation rules, and audit history.

- User and role management for Students, Supervisors, Coordinators, Panel Members, and Administrators.
- Semester and FYP-cycle configuration including group-size and workflow policies.
- Official short-proposal and long-proposal template management with versioned academic rules.
- Supervisor capacity, panel rubric, scoring criteria, and academic-deadline configuration.
- Deadline reminder/escalation rules and immutable audit history of administrative changes.

Primary Output: Academic governance configuration and deadline action rules.

Module 2: Student Team & Collaboration Hub

Description: This module manages student group formation and project collaboration. It supports AI-assisted teammate discovery for students who do not yet have a complete group and enforces the university rule that a student cannot belong to more than one active FYP group in the same cycle.

- FYP group creation, leader/member rules, group-size enforcement, and verified membership.
- Duplicate-group membership prevention using Student ID/Registration Number and current FYP cycle.
- AI teammate compatibility matching using 8–10 questions covering skills, domain, availability, working style, preferred role, and commitment.
- Join requests, invitations, accept/reject workflow, student-to-group matching, and temporary match chat.
- Seven collaboration channels (General, Proposal, Research, Supervisor Discussion, Tasks & Progress, Files & Resources, Announcements & Deadlines) plus a group AI chatbot.

Primary Output: Verified and compatible FYP team with a shared collaboration workspace.

Module 3: Student Proposal & AI Compliance Check

Description: This module owns the official proposal record and validates a student's short or long proposal against the university template and academic-quality rules. The same analysis can be viewed by authorized supervisors, coordinators, and panel members according to role permissions.

- Short/long proposal creation, upload, draft, submit, resubmit, and lifecycle-status management.
- Version history with AI-generated change summaries between proposal versions.
- University-format checking for font, font size, margins, alignment, line spacing, heading styles, page numbering, tables, and figures.
- Structural validation for required sections, section order, headings, references, appendices, and captions.
- Academic consistency analysis across problem statement, objectives, methodology, proposed solution, scope, and expected outcomes.

Primary Output: Official proposal version and Proposal Compliance & Readiness Report.

Module 4: Proposal Similarity & Originality Check

Description: This module compares a new proposal with archived FYPs to determine how much of the idea, problem, implementation scope, and methodology overlaps with previous work and where a genuine contribution can begin.

- Semantic embedding search against historical FYP records.
- Multi-level comparison of title, problem statement, objectives, modules/features, and methodology.
- Evidence showing the most related previous projects and their implemented scope.

- Contribution-boundary analysis separating inherited/overlapping work from genuinely new work.
- Gap and extension analysis based on previous limitations, missing features, and unfinished scope.

Primary Output: Originality, duplication, and contribution-boundary report.

Module 5: Supervisor Selection & Allocation

Description: This module identifies suitable supervisors and manages the complete assignment process using expertise relevance, research interests, current workload, capacity, and academic constraints.

- AI-based supervisor expertise matching using proposal content, research interests, and previous supervision history.
- Ranked supervisor recommendations with explainable match reasons and expertise gaps.
- Supervisor capacity and slot management showing total, filled, and available slots.
- Student request, supervisor accept/reject, duplicate-request protection, and final assignment workflow.
- Fair allocation and conflict checking using workload, preferences, capacity, and academic constraints.

Primary Output: Explainable supervisor recommendation and valid final assignment.

Module 6: Supervisor Proposal Review Dashboard

Description: After assignment, this module manages the formal supervisor-led proposal review process. Comments and requested changes remain linked to proposal versions so that the academic revision history is fully traceable.

- Paragraph-level, section-level, module-level, and full-proposal comments.
- Editing suggestions and formal revision requests from the assigned supervisor.
- Two-way student-supervisor discussion linked to the relevant proposal/revision item.
- Semantic comparison between old and revised proposal versions.
- AI revision-compliance checking that classifies requested changes as Resolved, Partially Resolved, or Unresolved while leaving final authority with the supervisor.

Primary Output: Revision Resolution Report and approved/revised proposal history.

Module 7: Coordinator Proposal Control

Description: This module gives the FYP Coordinator controlled academic routing over supervisor-approved proposals and provides visibility into delayed, incomplete, or workflow-inconsistent cases.

- Dedicated queue of supervisor-approved proposals ready for coordinator action.

- Hold, return-to-student, return-to-supervisor, and forward-to-panel actions.
- Version-aware, immutable history of coordinator decisions and reasons.
- Workflow bottleneck and abnormal-delay detection across proposal stages.
- Priority and conformance monitoring for urgent cases, skipped stages, unresolved requirements, and deadline risk.

Primary Output: Coordinator decision record and panel-ready proposal state.

Module 8: Panel Scheduling & Defense

Description: This module manages defense logistics after a proposal/project becomes eligible for panel evaluation. It allocates panel members, time slots, and rooms while preventing scheduling and academic conflicts.

- Panel-member assignment for eligible FYP groups.
- Availability management for panel members, supervisors, student groups, and rooms.
- Defense date, time, room, or online-session scheduling.
- Automatic detection of panel, supervisor, student, room, and conflict-of-interest clashes.
- Optimized scheduling and rescheduling to reduce conflicts and balance panel workload.

Primary Output: Conflict-free defense schedule and panel assignment.

Module 9: Panel Evaluation & Feedback

Description: This module manages formal panel evaluation using configured rubrics. It supports detailed feedback, AI-assisted defense preparation, evaluator-disagreement detection, and traceable feedback resolution.

- Criterion-based rubric scoring and recommendation by multiple panel members.
- Paragraph, section, module/feature, and overall proposal/project comments.
- AI-generated viva/defense questions based on proposal content and claimed contribution.
- Detection of significant evaluator disagreement in scores or assessment dimensions.
- Feedback-resolution workflow: Panel Comment $\rightarrow$ Coordinator Resolution $\rightarrow$ Student/Supervisor Revision $\rightarrow$ Panel Recheck, while preserving the original panel comment.

Primary Output: Structured panel evaluation, feedback history, and resolution status.

Module 10: Project Progress & Supervision Dashboard

Description: This module combines formal supervision evidence with milestone and task progress to present the current health of an active FYP and identify delay or delivery risks.

- Supervision meeting records, minutes, decisions, action items, owners, and due dates.
- AI extraction of tasks, owners, commitments, and deadlines from meeting notes.
- Milestone, task, completion-percentage, overdue-work, and scope-progress tracking.
- Project-health classification such as On Track, Attention Required, or Critical using recorded evidence.
- Risk explanation and recovery recommendations; when sufficient labelled data becomes available, predictive ML models and explainability can be added for delay probability estimation.

Primary Output: Project Progress, Action Compliance, Risk, and Recovery Report.

Module 11: Final FYP Archive & Handover

Description: This module creates the final institutional record of a completed FYP and prepares reusable knowledge for future students instead of treating the final report as a static file.

- Final project submission package containing report, proposal, source-code repository, dataset/model information, and supporting files.
- Automatic extraction of project metadata including domain, technologies, datasets, modules, and methodology.
- Classification of final scope as Completed, Partially Implemented, or Not Implemented.
- Extraction of limitations, unresolved issues, technical debt, and future work.
- Intelligent handover/continuation package containing reusable assets, dependencies, incomplete work, and recommended continuation points.

Primary Output: Final Project Archive Package and Continuation/Handover Package.

Module 12: FYP Knowledge Search & Analytics

Description: This module reads and reasons over archived institutional FYP knowledge. It provides evidence-based natural-language search for authorized users and produces institutional analytics without editing the underlying archive.

- Natural-language semantic search over archived FYP titles, abstracts, modules, technologies, datasets, outcomes, and limitations.
- RAG and GraphRAG-based question answering grounded in retrieved institutional evidence.
- Multi-hop queries across relationships such as Project $\rightarrow$ Technology $\rightarrow$ Dataset $\rightarrow$ Limitation $\rightarrow$ Supervisor.
- Source citations/evidence paths so generated answers can be traced to archived FYP records.

- Institutional analytics for domain trends, supervisor workload, approval turnaround, completion/delay patterns, panel statistics, and technology usage over time.

Primary Output: Evidence-Based FYP Knowledge Answers and Institutional Analytics Report.

10. Modules

Table 2 provides the module distribution and task allocation according to the four implementation checkpoints. The same pattern is used for each checkpoint: three modules are scheduled in P1 Mid, three in P1 Final, three in P2 Mid, and three in P2 Final.

Table 2: FYPHub Module Distribution and Task Allocation

Deadline	**Modules**	**Student**
P1 Mid	1: Admin Dashboard & Academic Management	Muhammad Maaz (23108179)
	2: Student Team & Collaboration Hub	Zain Ali Kiani (23108197)
	3: Student Proposal & AI Compliance Check	Danish Faraz (23108160)
P1 Final	4: Proposal Similarity & Originality Check	Muhammad Maaz (23108179)
	5: Supervisor Selection & Allocation	Zain Ali Kiani (23108197)
	6: Supervisor Proposal Review & Revision	Danish Faraz (23108160)
P2 Mid	7: Coordinator Proposal Control	Muhammad Maaz (23108179)
	8: Defense & Panel Scheduling	Zain Ali Kiani (23108197)
	9: Panel Evaluation & Feedback	Danish Faraz (23108160)
P2 Final	10: Project Progress & Supervision Dashboard	Muhammad Maaz (23108179)
	11: Final FYP Archive & Handover	Zain Ali Kiani (23108197)
	12: FYP Knowledge Search & Analytics	Danish Faraz (23108160)

11. Development Environment

Tools:	Visual Studio Code, Git/GitHub, PowerShell, Docker, Docker Compose, Swagger/OpenAPI, and Ollama for local AI model execution.
DBMS:	PostgreSQL with the pgvector extension for transactional academic data, proposal/version history, workflow records, archive metadata, and semantic vector search.
Platform:	Web-Based; Front-End: React + Vite + TypeScript with Tailwind CSS; Back-End: Python 3.12 with FastAPI for authenticated REST APIs, academic workflow rules, and AI service orchestration.
AI Environment:	Local and/or approved language models through Ollama, embedding models, RAG/GraphRAG pipelines, Python-based AI orchestration, machine-learning libraries for risk analysis, and optimization libraries for scheduling/allocation.

12. Significance

FYPHub is significant because it transforms the university's FYP process from a collection of disconnected administrative tasks into a connected academic intelligence ecosystem. The platform does not only digitize proposal forms; it links team formation, proposal quality, originality, supervision, coordinator routing, scheduling, panel evaluation, progress evidence, final archiving, and institutional knowledge reuse.

The main expected benefits are:

- **Improved Proposal Quality:** Students receive structured feedback on university formatting, missing sections, academic consistency, and readiness before formal review.
- **Reduced Duplication:** Semantic and functional comparison against previous FYPs reveals overlap and clarifies where genuine new contribution should begin.
- **Better Team Formation:** AI teammate matching and duplicate-membership rules support fair and compatible group creation.
- **Transparent Supervision:** Supervisor selection considers expertise and capacity, while revision comments and meeting actions remain traceable.
- **Efficient Academic Routing:** Coordinators can monitor proposal states, delays, and panel readiness from one controlled workflow.
- **Improved Defense Management:** Panel assignments and defense schedules are generated with conflict and availability checks.
- **Evidence-Based Progress Monitoring:** Meeting actions, milestones, tasks, and delays contribute to project-health and recovery recommendations.
- **Institutional Knowledge Reuse:** Final FYPs become searchable, reusable knowledge through RAG/GraphRAG and analytics rather than remaining isolated documents.

- **Human-Controlled AI:** AI supports analysis, explanation, retrieval, prediction, and recommendations, while authorized academic personnel retain final decision authority.

13. Tools and Techniques

FYPHub uses a modular web and AI architecture that separates deterministic academic rules from probabilistic AI assistance. The major tools and techniques are listed below.

- **Front-End:** React + Vite + TypeScript with Tailwind CSS will provide responsive role-specific interfaces for Student, Supervisor, Coordinator, Panel, and Administrator users [11].
- **Back-End:** Python 3.12 with FastAPI will provide authenticated REST APIs, role-based authorization, academic workflow rules, AI-service endpoints, and integration logic [12].
- **Database:** PostgreSQL will store users, teams, proposals, versions, assignments, reviews, coordinator actions, schedules, evaluations, progress records, and archive metadata [13]. The pgvector extension will support proposal embeddings, semantic retrieval, and historical FYP similarity search [14].
- **AI and Intelligence:** LLMs and embeddings will support proposal-quality analysis, semantic revision checking, teammate explanations, viva-question generation, summarization, and semantic search. RAG will ground generated answers in retrieved institutional evidence [3], while GraphRAG will add graph-structured retrieval for broader and multi-hop reasoning across archived FYP relationships [4]. Optimization techniques will support supervisor allocation and defense scheduling, while project-risk analysis will begin with deterministic/hybrid evidence and may later use supervised ML when sufficient labelled data becomes available.
- **Security and Academic Rules:** Role-Based Access Control (RBAC) and deterministic backend/database constraints will enforce group uniqueness, supervisor slots, proposal-version eligibility, coordinator routing, panel permissions, and administrative policies rather than delegating these controls to an LLM.
- **Development Tools:** Docker and Docker Compose will support reproducible infrastructure [15]; Git/GitHub will support collaborative version control; Swagger/OpenAPI will support API documentation and testing; Ollama will support local AI model execution.

Overall Workflow: Admin Setup → Team Formation → Proposal Submission & AI Compliance → Similarity & Originality Check → Supervisor Selection & Allocation → Supervisor Review & Revision → Coordinator Proposal Control → Defense & Panel Scheduling → Panel Evaluation & Feedback → Project Progress & Supervision → Final FYP Archive & Handover → FYP Knowledge Search & Analytics.

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